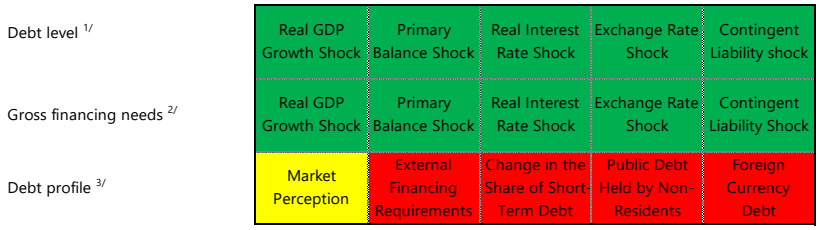
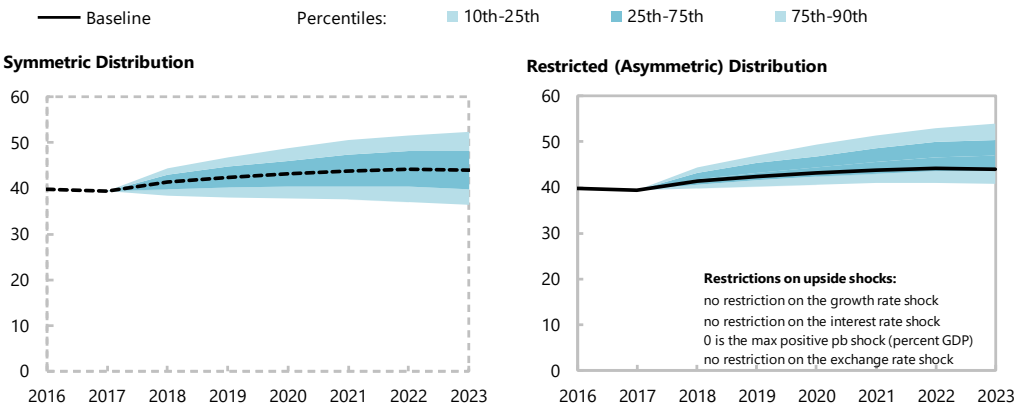
Figure 5. FYR Macedonia: General Government DSA Risk Assessment

Heat Map



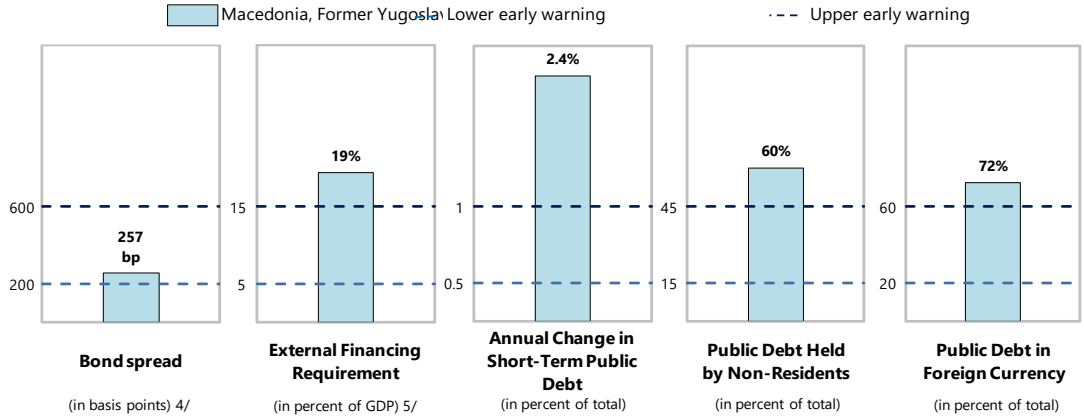
Evolution of Predictive Densities of Gross Nominal Public Debt

(in percent of GDP)



Debt Profile Vulnerabilities

(Indicators vis-à-vis risk assessment benchmarks, in 2017)



Source: IMF staff.

1/ The cell is highlighted in green if debt burden benchmark of 70% is not exceeded under the specific shock or baseline, yellow if exceeded under specific shock but not baseline, red if benchmark is exceeded under baseline, white if stress test is not relevant.

2/ The cell is highlighted in green if gross financing needs benchmark of 15% is not exceeded under the specific shock or baseline, yellow if exceeded under specific shock but not baseline, red if benchmark is exceeded under baseline, white if stress test is not relevant.

3/ The cell is highlighted in green if country value is less than the lower risk-assessment benchmark, red if country value exceeds the upper risk-assessment benchmark, yellow if country value is between the lower and upper risk-assessment benchmarks. If data are unavailable or indicator is not relevant, cell is white.
Lower and upper risk-assessment benchmarks are:

200 and 600 basis points for bond spreads; 5 and 15 percent of GDP for external financing requirement; 0.5 and 1 percent for change in the share of short-term debt; 15 and 45 percent for the public debt held by non-residents; and 20 and 60 percent for the share of foreign-currency denominated debt.

4/ Long-term bond spread over German bonds, an average over the last 3 months, 12-Jul-18 through 10-Oct-18.

5/ External financing requirement is defined as the sum of current account deficit, amortization of medium and long-term total external debt, and short-term total external debt at the end of previous period.

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Figure 6. FYR Macedonia: Public Sector General Government Debt Sustainability (DSA) – Baseline Scenario

(In percent of GDP unless otherwise indicated)

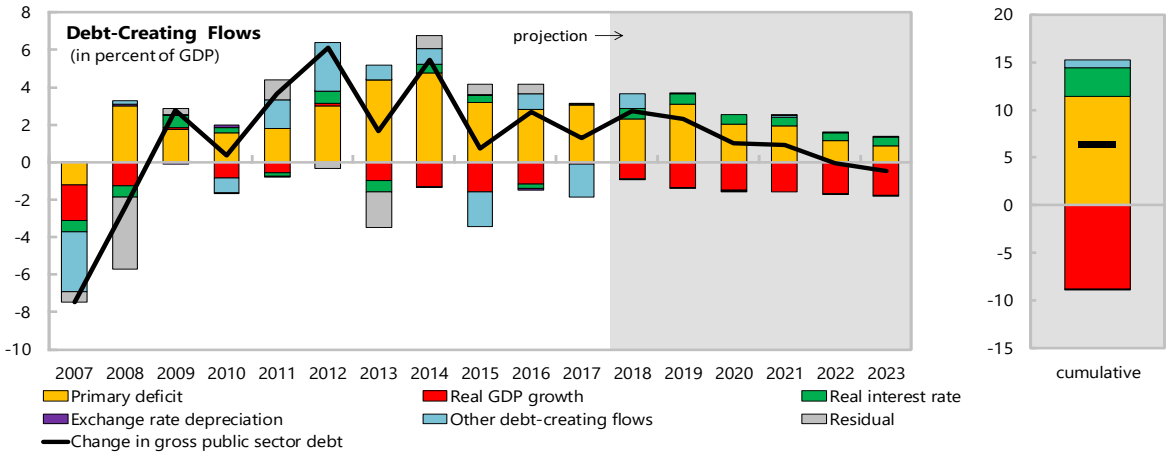
Debt, Economic and Market Indicators [1/]

	Actual			Projections					
	2007-2015 [2/]	2016	2017	2018	2019	2020	2021	2022	2023
Nominal gross public debt	32.5	46.7	48.0	50.7	53.0	54.0	54.9	54.9	54.4
Public gross financing needs	16.4	16.6	14.9	15.4	14.4	14.2	14.9	10.7	11.1
Real GDP growth (in percent)	3.0	2.8	0.2	2.0	2.8	3.0	3.1	3.3	3.4
Inflation (GDP deflator, in percent)	2.8	3.5	3.4	1.7	2.0	2.0	2.2	2.2	2.2
Nominal GDP growth (in percent)	5.9	6.4	3.7	3.7	4.9	5.1	5.4	5.6	5.7
Effective interest rate (in percent) [4/]	3.1	3.0	3.5	3.0	3.2	3.2	3.2	3.1	3.1

As of October 10, 2018		
Sovereign Spreads		
EMBIG (bp) 3/	293	
5Y CDS (bp)	n.a.	
Ratings	Foreign	Local
Moody's	n.a.	n.a.
S&Ps	BB-	BB-
Fitch	BB	BB

Contribution to Changes in Public Debt

	Actual			Projections							
	2007-2015	2016	2017	2018	2019	2020	2021	2022	2023	cumulative	debt-stabilizing primary balance [9/]
Change in gross public sector debt	1.2	2.7	1.3	2.7	2.3	1.0	0.9	-0.1	-0.5	6.4	
Identified debt-creating flows	1.7	2.2	1.2	2.7	2.3	1.0	0.9	-0.1	-0.5	6.4	
Primary deficit	2.5	2.8	3.1	2.3	3.1	2.0	1.9	1.2	0.9	11.4	-1.4
Primary (noninterest) revenue and gra	29.9	28.4	29.1	29.7	30.9	30.9	30.8	30.8	30.8	183.9	
Primary (noninterest) expenditure	32.4	31.2	32.2	32.0	34.0	32.9	32.8	32.0	31.7	195.3	
Automatic debt dynamics [5/]	-0.8	-1.5	-0.1	-0.3	-0.8	-1.0	-1.1	-1.3	-1.3	-5.8	
Interest rate/growth differential [6/]	-0.9	-1.4	-0.1	-0.3	-0.8	-1.0	-1.1	-1.3	-1.3	-5.8	
Of which: real interest rate	0.1	-0.2	0.0	0.6	0.6	0.5	0.5	0.4	0.4	3.0	
Of which: real GDP growth	-0.9	-1.2	-0.1	-0.9	-1.4	-1.5	-1.6	-1.7	-1.8	-8.8	
Exchange rate depreciation [7/]	0.0	-0.1	0.0	...	...	...	...	...	...	...	
Other identified debt-creating flows	0.0	0.9	-1.8	0.8	0.0	0.0	0.1	0.0	0.0	0.8	
Privatization receipts (negative)	-0.2	0.0	-0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
Contingent liabilities	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
Changes in cash, deposits, and securi	0.2	0.9	-1.7	0.8	0.0	0.0	0.1	0.0	0.0	0.8	
Residual, including asset changes [8/]	-0.4	0.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	



Source: IMF staff.

1/ Includes general government and SOEs debt.

2/ Based on available data.

3/ Long-term bond spread over German bonds.

4/ Defined as interest payments divided by debt stock (excluding guarantees) at the end of previous year.

5/ Derived as [(r - π(1+g) - g + ae(1+r)]/(1+g+π+gπ)) times previous period debt ratio, with r = interest rate; π = growth rate of GDP deflator; g = real GDP growth rate; a = share of foreign-currency denominated debt; and e = nominal exchange rate depreciation (measured by increase in local currency value of U.S. dollar).

6/ The real interest rate contribution is derived from the numerator in footnote 5 as r - π (1+g) and the real growth contribution as -g.

7/ The exchange rate contribution is derived from the numerator in footnote 5 as ae(1+r).

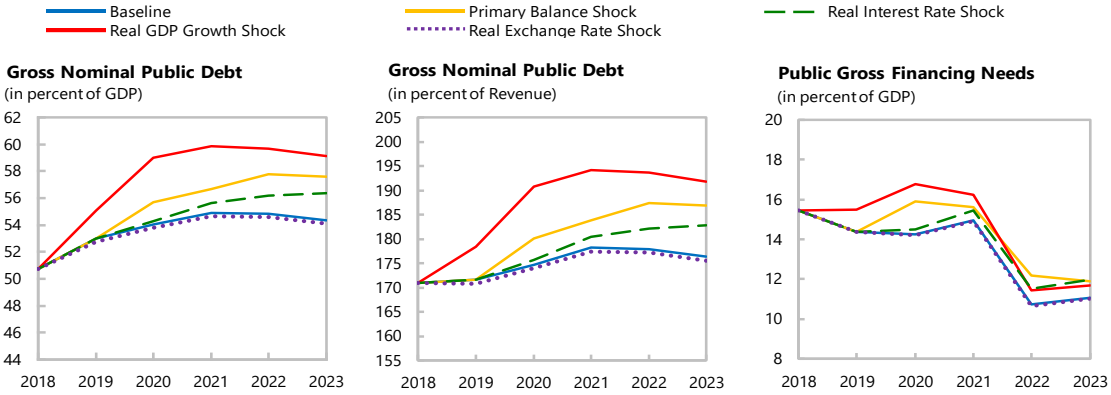
8/ Includes asset changes and interest revenues (if any). For projections, includes exchange rate changes during the projection period.

9/ Assumes that key variables (real GDP growth, real interest rate, and other identified debt-creating flows) remain at the level of the last projection year.

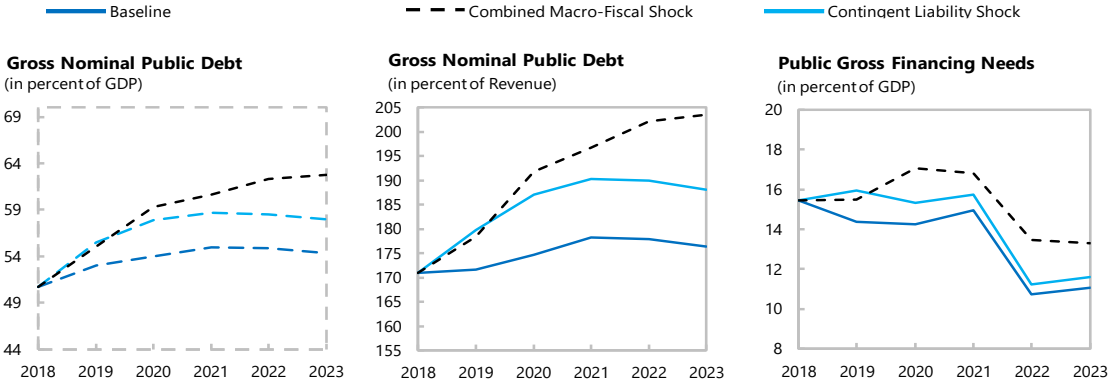
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Figure 7. FYR Macedonia: Public Sector DSA – Stress Tests

Macro-Fiscal Stress Tests



Additional Stress Tests



Underlying Assumptions

(in percent)

Primary Balance Shock	2018	2019	2020	2021	2022	2023
Real GDP growth	2.0	2.8	3.0	3.1	3.3	3.4
Inflation	1.7	2.0	2.0	2.2	2.2	2.2
Primary balance	-2.3	-3.1	-3.7	-2.0	-2.3	-1.3
Effective interest rate	3.0	3.2	3.2	3.3	3.2	3.2
Real Interest Rate Shock						
Real GDP growth	2.0	2.8	3.0	3.1	3.3	3.4
Inflation	1.7	2.0	2.0	2.2	2.2	2.2
Primary balance	-2.3	-3.1	-2.0	-1.9	-1.2	-0.9
Effective interest rate	3.0	3.2	3.7	4.0	4.3	4.5
Combined Shock						
Real GDP growth	2.0	0.8	1.0	3.1	3.3	3.4
Inflation	1.7	1.5	1.5	2.2	2.2	2.2
Primary balance	-2.3	-3.9	-3.7	-2.0	-2.3	-1.3
Effective interest rate	3.0	3.2	3.7	4.0	4.3	4.4

Real GDP Growth Shock	2018	2019	2020	2021	2022	2023
Real GDP growth	2.0	0.8	1.0	3.1	3.3	3.4
Inflation	1.7	1.5	1.5	2.2	2.2	2.2
Primary balance	-2.3	-3.9	-3.6	-1.9	-1.2	-0.9
Effective interest rate	3.0	3.2	3.2	3.3	3.2	3.2
Real Exchange Rate Shock						
Real GDP growth	2.0	2.8	3.0	3.1	3.3	3.4
Inflation	1.7	1.7	2.0	2.2	2.2	2.2
Primary balance	-2.3	-3.1	-2.0	-1.9	-1.2	-0.9
Effective interest rate	3.0	3.2	3.2	3.2	3.1	3.1
Contingent Liability Shock						
Real GDP growth	2.0	0.8	1.0	3.1	3.3	3.4
Inflation	1.7	1.5	1.5	2.2	2.2	2.2
Primary balance	-2.3	-4.3	-2.0	-1.9	-1.2	-0.9
Effective interest rate	3.0	3.3	3.2	3.2	3.2	3.2

Source: IMF staff.

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